



Fig. 2: Raspberry Pi desktop screen

my television screen, I was mesmerised.

To make Raspberry Pi open the GUI screen every time it boots up, all you have to do is to make a change in the `raspi-config` page. Select Sudo Raspi-config and bring the cursor down on Automatic GUI Boot Up. Select → Finish → Reboot. And, it's done.

Connecting the USB camera and Wi-Fi dongle. Connect the Wi-Fi adapter and webcam to USB slots. You will run short of USB ports now because Raspi-B model has only two USB ports. Find a USB hub with multiple USB ports and grab a hub that has a small, 5V power supply input to it.

(Remember, wireless needs more power. Therefore, if the hub is not powered by a suitable 5V supply, the Raspberry Pi may malfunction.)

Power-on the USB hub; connect the wireless adapter (Wi-Fi dongle), web camera, keyboard and mouse to the USB hub. Go to Raspberry Pi desktop and double-click the wireless module (Wi-Fi Config). Most wireless adapters are easily detected and installed on Raspberry Pi.

Set up a Wi-Fi connection. After double clicking Wi-Fi Config icon, the `wpa_gui` page will open up. Click Search and a list of available wireless connections will come up. Select a connection and enter the IP address, if required, or leave it as it is for DHCP mode, select the authentication mode (mostly WEP) from the pull-down menu and then enter WEP key in the first blank space. Click Fin-

ish and Connect. The symbol will turn to Connected.

(For connecting to a mobile hotspot like cell phones, the WPA2 psk type authentication is used. Select the same from the pull-down menu.)

Checking network connection. To check whether your Wi-Fi is connected to the network or not, you need to go to Terminal and issue ping command as given below.

```
pi@raspberrypi:/$ ping raspberrypi.org
```

If you get continuous ping response, it means you are online and connected. Otherwise, you need to check the connections and, if required, need to go through the steps below.

Enter the following command at Raspberry Pi terminal:

```
pi@raspberrypi:/$ dmesg | grep ^usb
```

This command will furnish details of the kernel ring buffer of all the devices installed, matching with the word USB. Look out for 802.11 and the manufacturer's name (realtek or ralink) associated with it. This means that the hardware is built on that chip and Raspberry Pi needs the driver (say realtek) for it. (On the latest Raspbian OS, this step is not required because it finds the suitable driver automatically.)

Driver installation. You can issue the following command to search the driver on your Raspberry Pi computer:

```
bera@Acer-AOD255E:~/Desktop/Raspberrypi/OS$ sudo apt-cache search realtek
```

The command will give details of the packages available for that manufacturer. Since your Raspberry Pi is still not connected to the Internet, you need to go to a Linux PC that has an Internet connection and search for realtek drivers online. Go to the manufacturer's website and download Debian wireless package (e.g., `firmware-realtek.deb`, `firmware-ralink.deb`) from their website. Copy this package in the `/home/pi` folder of the SD card and issue the following command to the Raspberry Pi:

```
pi@raspberrypi:/$ sudo apt-get install firmware-realtek.deb
```

This will install the right firmware on Raspberry Pi. Once the firmware is installed, disconnect the wireless adapter and reconnect it to Raspberry Pi. This will restart the kernel search process for the right software driver.

With the firmware installed, setting-up the connection is pretty straightforward now. Go back to the desktop and double-click WiFi Config icon and follow the steps mentioned above under Set-Up a Wi-Fi Connection section.

Setting up the webcam server.

`ffmpeg` has some issues on Raspberry Pi as a result of which `ffserver` does not work easily. Some staunch Raspberry Pi supporter claims to tweak it to work but I find it very complicated and not worth when various other easy choices are available.

Apache2 server with webcam software makes our job very easy but, before that, to ensure the webcam is working, we have to install VLC software. Along with showing webcam images, VLC can play all kinds of media files including *.flv files, which was not possible in the earlier OS.

Installation of VLC Apache2. To install it, issue the following command:

```
pi@raspberrypi:/$ sudo apt-get install vlc apache2 webcam
```

This will install the wav trio—webcam, Apache2 and VLC media player. Go to the desktop, in the media section, click on VLC media player. Go to File → Capture Device, and see if the camera images appear or not. VLC is smart enough to find your first camera at `/dev/video0` and the video driver as `video4linux2` (if not, select from the pull-down menu). It will take some time for the images to appear on the screen. The images may appear broken at many places, but do not worry about it now.

Webcam software has no default configuration file to run with, so we have to make one now using Nano editor. Nano is the most beautiful,